

Understanding the dataset :

Part 1 :

dataset consists of 1,370 knee MRI exams It contains

1,104 (80.6%) abnormal exams

319 (23.3%) ACL tears

508 (37.1%) meniscal tears

There are **3 separate label files**, and each one independently says Positive (1) or Negative (0) for that condition:

Exam ID	abnormal	acl	meniscu s
0000	1	0	1
0001	1	1	0
0002	0	0	0
0003	1	1	1

So exam 0000 above is: **has** abnormal , **No** ACL tear , **has** meniscal tear

And exam 0002 is a **completely normal knee** — negative on all three.

Part 2 :

A patient CAN be:

abnormal=1, acl=1, meniscus=0 → ACL tear only

abnormal=1, acl=0, meniscus=1 → Meniscal tear only

abnormal=1, acl=1, meniscus=1 → BOTH tears at once ← this happens!

abnormal=1, acl=0, meniscus=0 → Abnormal but neither specific tear

abnormal=0, acl=0, meniscus=0 → Completely healthy

Notice also that **if acl=1 or meniscus=1, then abnormal is always 1** — you can't have a tear without being abnormal. But abnormal=1 doesn't guarantee either specific tear.